SQE Assignment#2

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For IT companies do some research and list top 5 testing types which are in place. For each type mention the tools that help in automation of that type. Also search for the AI agents that helps the most in doing that type of test.

1. Unit testing

Validates the smallest testable units of source codes - such as individual functions, methods, or classes - in strict isolation from external dependencies.

- Automation Tools and Frameworks: JUnit 5 (Java), PyTest (Python), Jest/Vitest (JavaScript/TypeScript), NUnit (.NET), Moq/Jest (mocking/stubbing).
- Leading AI Agents for doing:
  - Diffblue Cover: An autonomous generative AI agent specifically built for enterprise codebases (predominantly Java) that analyses code structure and independently writes, validates and maintains 100% executable regression and unit suites.
  - Qodo (formerly Codium AI): An agentic code-integrity platform that operates inside IDE's and CI/CD pipelines to analyse edge cases, logic branches and boundary conditions, drafting comprehensive unit test suites with required mocks.

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2. Integration and API testing

This layer focuses on interactions across subsystem boundaries, database transactions, microservice contracts, and REST/GraphQL endpoint behaviors without relying on a full graphical interface.

- Automation Tools and frameworks: Postman / Newman CLI, REST Assured, Karate DSL, supertest, Pact (contract testing), Wiremock.
- Leading AI agents for Test Designing:
  - KushoAI: An autonomous agent that reads OpenAPI / Swagger specifications, Postman collections, or curl traces to map parameter dependencies, generate combinatorial payload variations, and build exhaustive functional API assertions.
  - Keploy: An AI testing Agent that passively intercepts real distributed network traffic (HTTP, gRPC, DB calls) in staging/dev and synthesizes deterministic integration test cases and automated mocks without manual script authoring.

3. End-to-End (E2E) & UI Regression Testing:

E2E testing replicates complete user workflows across browsers and client interfaces, confirming that the entire application stack - front-end, back-end services, network and database - functions seamlessly together.

- Automation Tools and frameworks: Playwright, Cypress, Selenium WebDriver, Appium (mobile), Puppeteer.
- Leading AI agents for Test Designing:
  - Virtuoso QA (Step IQ / Generator): Autonomous

agent that parses requirements (PRDs, Jira tickets, or Checkin files) into functional UI test journeys. It relies on computer vision and DOM-state heuristic for self-healing locators when the UI structure changes.

- Kane AI (by TestMu / LambdaTest): A GenAI-native testing agent capable of planning, executing, and repairing complex web and mobile user journeys directly from natural language specifications or recorded sessions.

4. Performance and load Testing

Performance testing evaluates system stability, throughput & latency and resource consumption under peak, sustained or stress loads.

- Automation Tools and frameworks: Grafana k6 (Java-Script-based). Apache JMeter, Gatling (Scala/Java) locust (Python-based distributed execution)
- Leading AI agents for Test Devising:
  - Akamas: An autonomous performance optimization agent that uses reinforcement learning to formulate stress-testing parameters, execute load simulations evaluate bottleneck metrics and recommend optimal cloud/JVM configuration
  - LoadRunner Cloud AI (Micro Focus / OpenText): Uses predictive analytics and adaptive workload generation agents to synthesize dynamic user concurrency curves on production APM logs (eg, DataDog, Dynatrace).

5. Security Testing (SAST, DAST and PenTesting):

Security testing systematically checks applications for vulnerabilities, misconfigurations, injection flaws and adherence to security baselines (such as the OWASP top 10).

- Automation Tools and frameworks: OWASP ZAP, Burp suite Enterprise, Snyk, SonarQube, Semgrep, Trivy (Container security)
- Leading AI agents for Bot Detection:
- Snyk DeepCode and Semgroup AI: Specialized static analysis agents that trace tainted data flow paths across multi-file architectures to model vulnerability vectors and automatically propose fix-validation tests.
- Tassiel/Hexway Pentest Agents: Agentic penetration testing tools that autonomously map application attack surfaces, craft context-aware payload injections (eg prompt injections, SSRF, SQLi) and validate exploitability in CI pipelines.